

Digital Logic Circuits (Spring 2026)

HW #1 (Due: March 25, 2026)

19. What are two ways of representing zero in 1's complement form?
20. How is zero represented in 2's complement form?
24. Express each decimal number as an 8-bit number in the 1's complement form:
(a) -34 (b) $+57$ (c) -99 (d) $+115$
25. Express each decimal number as an 8-bit number in the 2's complement form:
(a) $+12$ (b) -68 (c) $+101$ (d) -125
29. Express each of the following sign-magnitude binary numbers in single-precision floating-point format:
(a) 0111110000101011 (b) 100110000011000
30. Determine the values of the following single-precision floating-point numbers:
(a) 1 10000001 010010011 10001000000000
(b) 0 11001100 100001111101001000000000
33. Perform each addition in the 2's complement form:
(a) $10001100 + 00111001$ (b) $11011001 + 11100111$
34. Perform each subtraction in the 2's complement form:
(a) $00110011 - 00010000$ (b) $01100101 - 11101000$
35. Multiply 01101010 by 11110001 in the 2's complement form.

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19. $+0: 00000000$
 $-0: 11111111$

20. only: 00000000

24. (a) $(34)_{10} = (00100010)_2$
 $-34: 1's \text{ complement} = 11011101$

(b) $(57)_{10} = (00111001)_2$
 $+57: 1's \text{ complement} = 00111001$

(c) $(99)_{10} = (01100011)_2$
 $-99: 1's \text{ complement} = 10011100$

(d) $(15)_{10} = (01110011)_2$
 $+15: 1's \text{ complement} = 01110011$

25. (a) $(12)_{10} = (00001100)_2$
 $+12: 2's \text{ complement} = 00001100$

(b) $(68)_{10} = (01000100)_2$
 $-68: 2's \text{ complement} = 10111100$

(c) $(101)_{10} = (01100101)_2$
 $+101: 2's \text{ complement} = 01100101$

(d) $(125)_{10} = (01111101)_2$
 $-125: 2's \text{ complement} = 10000011$

29. (a) $0111, 10000101011 = X$
 $1.11110000101011 \times 2^{14}$

$E = 14 + 17 = (31)_{10} = (0001101)_2$

$X = 01000110111100001010110000000000$

29. (b) $100, 11000001, 1000 = y$

$y = -1.10000011000 \times 2^{11}$

$E = 11 + 17 = (28)_{10} = (10001010)_2$

$y = 11000101010000011000000000000000$

30 (a) $11000001010010011100010000000000$
 $(10000001)_2 = 129 = 127 + 2$

$-1.01001001110001 \times 2^2 = -101.001001110001$

(b) $01100110010000111101001000000000$
 $11001100_2 = 204 = 127 + 77$
 $1.100001111101001 \times 2^{77}$

33. (a) 10001100
 $+00111001$
 $\hline 11000101$

(b) 11011001
 $+11100111$
 $\hline 11000000$

34. (a) $00110011 - 00010000$
 $= 00110011 + 11101111 + 1 = 00100011$

(b) $01100101 - 11101000$
 $= 01100101 + 00011000 = 01111101$

10010110

35. 01101010×11110001

By Booth:

$= 01101010$
 $000-1001-1$

$\hline$
 1111111110010110
 00000000101010
 00000000000000
 00000000000000
 11110010110
 000000000000
 000000000000
 00000000
 $\hline 1111100111001010$

$= 1111100111001010$ (2's complement)